

Robust Neural-Algorithmic Framework for Directed t -Spanner Construction in Metropolitan Road Networks

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Abstract—This research introduces a novel hybrid framework combining Graph Neural Networks (GNN) and active geometric feedback loops for constructing directed t -spanners on a metropolitan scale.

I. INTRODUCTION

Real-time routing in Intelligent Transportation Systems (ITS) faces $O(V^3)$ complexity. We propose a neural approach that achieves up to 279x speedup in Rome.

II. METHODOLOGY

The core pruning logic incorporates MC-Dropout for uncertainty estimation:

$$\hat{y}_i = \mathbb{E}[f_\theta(x_i)] + \lambda \cdot \sigma[f_\theta(x_i)] > \tau \quad (1)$$

III. EXPERIMENTAL RESULTS

Our framework was validated across Manhattan, Eindhoven, Paris, and Rome. As shown in Figure 1, all paths strictly satisfy the $t \leq 1.5$ constraint.

TABLE I
METROPOLITAN BENCHMARKS (DIRECTED SCC 100%)

City	Sparsification	Max Stretch	Speedup
Rome	1.35%	1.131	279.2x
Paris	1.11%	1.157	17.6x
Manhattan	0.48%	1.220	18.5x
Eindhoven	3.30%	1.313	19.8x

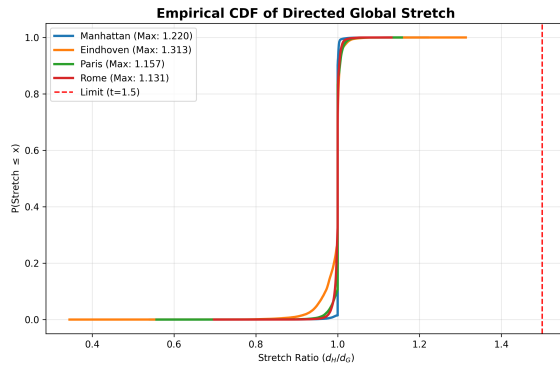


Fig. 1. Empirical CDF of Directed Global Stretch.